Depression Severity Screening in Students Using Machine Learning

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ABSTRACT

Depression is a major mental health problem among university students and can have negative consequences for academic performance, social relationships and overall well-being. Recognizing the signs of depression early is important for timely intervention and support. In this study, we propose a web-based platform for Depression Severity Screening which combines standardized mental health assessments with machine learning algorithms for predicting the level of depression severity. The system uses student mental health survey data along with demographic, behavioral, and psychological characteristics. Four feature-set strategies have been evaluated using Logistic Regression, Support Vector Machine (SVM) and Random Forest classifiers. The experimental results showed that the direct symptom-based features were the most predictive ones while the correlation-based feature selection improved the model efficiency by eliminating the irrelevant information. The best models achieved an accuracy of up to 88% for the classification of the severity of depression. we developed a full-stack web application using Next.js, Flask and MongoDB, allowing for practical deployment. The platform includes secure user authentication, online assessment, real-time prediction generation, assessment history tracking, and recommendation support. The results demonstrate that machine learning and modern web technologies can be effectively integrated to create an accessible decision support tool for screening depression severity in students. This system is intended to assist with early awareness and screening and is not a replacement for diagnosis or treatment by a professional mental health provider.

keywords: Depression Severity Screening, Student Mental Health, Machine Learning, PHQ-9, Logistic Regression, Support Vector Machine, Random Forest, Web Application, Mental Health Assessment, Decision Support System.

Chapter 1

Introduction

1.1 Background: Artificial Intelligence in Mental Health

Worldwide, mental health disorders are a growing problem for public health. Depression is one of the most common and debilitating disorders. Depression can have negative effects on emotional health, academic achievement, productivity, social relationships and life satisfaction. University students have unique vulnerabilities associated with academic stress, social transition, financial concern and lifestyle change associated with higher education.

Depression is a major health problem all over the world and is considered by the World Health Organization (WHO) as one of the leading causes of disability worldwide, affecting hundreds of millions of people [1]. Recent studies have also reported increasing rates of depressive symptoms among university students highlighting the need for effective screening and early intervention strategies [4].

In the past, mental health assessments were conducted by trained professionals through clinical interviews and questionnaires. While these techniques have been clinically validated, they are time and resource intensive and may not be available to large numbers of students.

New tools, based on recent advances in Artificial Intelligence (AI) and Machine Learning (ML), can support the assessment of mental health and the early detection of risk. Machine learning algorithms can handle large amount of structured data of mental health and find patterns with respect to different levels of severity of depression . These can support the development of scalable and accessible screening systems, together with standardized psychometric tools. However, these systems are not designed to replace professional diagnosis or treatment but are rather meant to support awareness and early detection.

1.2 Student-Focused Screening

Machine learning has been applied to several mental health conditions including depression, anxiety disorders, stress-related issues and prediction of suicide risk . Predictive models have been developed using various sources of data including clinical records, social media activity, speech, wearable sensors, behavioural indicators and questionnaire responses [7].

In this regard, one of the most relevant applications is the screening for depression among university students, since mental health problems are becoming more common in academic environments. Preventive mental health interventions are a relevant target for student populations since they are often under pressure regarding their academic performance, career uncertainty, social relationships, and personal development.

Standardized instruments for the assessment of depression, e.g. the Patient Health Questionnaire-9 (PHQ-9), the Beck Depression Inventory-II (BDI-II) and the Center for Epidemiologic Studies Depression Scale (CES-D), provide systematic measures of the intensity and severity of depressive symptoms. These instruments provide measurable data that can be used by machine learning algorithms to help in the screening and severity prediction of depression.

The study focused on student mental health and aimed to apply machine learning models to predict depression severity based on standardized assessment data. The study is not only about the predictive modelling but also integrating such models into a practical web based screening platform.

1.3 Challenges in Traditional Mental Health Screening

Traditional mental health screening methods primarily rely on clinical interviews, psychological assessments, and professional evaluation. Although these approaches are widely accepted and clinically validated, they face several practical limitations. Access to qualified mental health professionals is often limited, particularly in educational institutions with large student populations. As a result, many students experiencing depressive symptoms may not receive timely assessment or support.

Furthermore, conventional screening procedures can be time-consuming and resource-intensive, making large-scale implementation difficult. Many students are also reluctant to seek professional help due to social stigma, fear of judgment, or concerns about privacy. These barriers can delay identification of mental health issues and reduce the effectiveness of early intervention efforts.

Given the growing prevalence of depression among students, there is a need for scalable, accessible, and cost-effective screening solutions. Technology-driven approaches, particularly those based on machine learning, have the potential to support early detection by analyzing assessment data efficiently and assisting in the identification of individuals who may be at risk.

Chapter 2

Literature Survey and Related Work

2.1 PHQ-9 as a Depression Screening Instrument

The Patient Health Questionnaire-9 (PHQ-9) is a widely used screening, diagnostic and severity measure for depressive symptoms. Kroenke et al. developed the PHQ-9, a reliable and valid scale with high internal consistency, test-retest reliability, and criterion validity in clinical and non-clinical samples [3]. The nine items on the PHQ-9 directly correspond to the diagnostic criteria of major depressive disorder and hence the PHQ-9 is an appropriate scale to be used for screening and measuring the severity of depression.

Besides its empirically proven validity, the PHQ-9 is institutionally supported and recommended by major health institutions. The PHQ-9 is recommended by World Health Organization (WHO) and American Psychiatric Association (APA) for screening of depression and to assess the severity of depression. The Ministry of Health and Family Welfare, Government of India, has officially approved the use of PHQ-9 for public health and primary health care programs.

The PHQ-9's most distinctive feature is its ordinal severity scale, in which cumulative scores are assigned to increasingly severe levels of depression symptoms and range from minimal to severe [3]. This enables the PHQ-9 to maintain meaningful information and also be amenable to statistical analysis and machine learning classification of severity levels. Consequently, the PHQ-9 is often used as the target variable in machine learning based depression screening studies. Many of these approaches do not fully exploit the ordinal nature of the PHQ-9, collapsing levels of severity into nominal or binary categories.

2.2 BDI-II as a Depression Screening Instrument

One of the most frequently used self-report instruments to evaluate the presence and severity of depressive symptoms is the Beck Depression Inventory-II (BDI-II). The instrument measures different aspects of depression including emotional, cognitive, motivational and physical symptoms thus enabling a complete assessment of an individual's psychological state [5].

The BDI-II is a series of items that assess the severity of depressive symptoms experienced over a recent period. Responses are scored numerically and summed to give a total score which is then assigned to different levels of depression severity. This scoring system allows the BDI-II to be used as a screening tool as well as a measure of symptom severity [5].

One important strength of the BDI-II is its wide application in clinical practice and research. The instrument has shown good psychometric properties and has been widely used for assessing depressive symptoms in different populations [5]. The BDI-II is widely used as a reference standard in studies for depression assessment and mental health evaluation [4], [5], as it covers a broad spectrum of psychological and behavioral manifestations of depression.

BDI-II has also defined severity categories which makes it applicable for machine learning applications. BDI-II scores can be used as target variables in predictive modeling studies to classify depression severity levels. Similarly to PHQ-9-based approaches, many machine learning studies dichotomize or nominalize BDI-II outcomes, possibly neglecting the ordinal relationship between severity levels. Keeping the ordinal nature of BDI-II severity classes might increase interpretability and clinical relevance of prediction models.

2.3 CES-D as a Depression Screening Instrument

The Center for Epidemiologic Studies Depression Scale (CES-D) is a self-report scale developed to measure depressive symptomatology in the general population. Unlike diagnostic tools that are primarily designed for clinical work, the CES-D has been developed for large scale screening and epidemiological studies and can thus be especially useful in identifying individuals who are likely to have significant depressive symptomatology [4].

The CES-D was designed to measure several symptoms of depression, such as depressed mood, hopelessness, sleep disturbances, loss of appetite and problems with daily functioning. Responses to individual items are scored and summed to yield a total score of severity of depressive symptoms. Higher scores indicate higher likelihood of clinically significant depressive symptoms [4].

The main strength of the CES-D is that it is applicable to community and population surveys of mental health . The instrument has been used extensively in studies with different demographic groups, and has demonstrated acceptable reliability and validity for depression screening [4]. It is widely used in public health and behavioral health research because it can measure depressive symptoms in large populations efficiently.

The CES-D scale is scored quantitatively and is therefore amenable to machine learning applications. In predictive modeling studies, CES-D scores can be used to classify individuals with different levels of severity of depressive symptoms. Many machine learning approaches such as PHQ-9 and BDI-II combine the CES-D results into binary or nominal categories, which can reduce the information content of the original severity scale. The graded CES-D scores might contain more information useful for better predicting depression with higher accuracy and interpretability.

2.4 Machine Learning in Mental Health Research

The machine learning techniques have been widely applied in mental health research, especially in automatic detection and assessment of depressive disorders. Mao et al. in a systematic review for depression prediction and classification identified several algorithms including Logistic Regression (LR), Support Vector Machines (SVM), Decision Trees (DT), Random Forests (RF) and Deep Learning (DL) models [7]. These approaches have been developed using a variety of data sources, including clinical interviews, speech characteristics, textual content, facial expressions, physiological measurements, and self-reported questionnaire responses [7].

The review suggests that most of the studies focus mainly on predictive performance and evaluation is often done based on accuracy, precision, recall and f1-score [7]. Therefore, a lot of research effort has been invested in the development of more and more complex models with higher classification performance.

However, such advances have improved the predictive accuracy of the models, model interpretability, transparency and explainability [7] has been relatively less attended to. Many state-of-the-art models are black box systems, providing little insight into what factors drive the prediction or the relationship between input features and depression outcomes. This lack of transparency presents a barrier to clinical uptake where the reasons for a prediction are often as important as the prediction itself.

Thus, recent research has focused on the development of machine learning models that not only predict but also provide interpretable and clinically meaningful explanations. Such approaches can increase trust, enable validation by healthcare professionals, and facilitate the responsible deployment of artificial intelligence systems in mental health screening applications [7, 11].

2.5 Existing ML-Based Depression Detection Systems

Existing models for depression prediction have mainly been concerned with predictive accuracy, and black-box models like deep learning and ensemble methods have been widely used. Many of these models are non-transparent about the influence of individual features on the prediction and very few studies try to align the prediction with clinical interpretation as mentioned by Mao et al. [7]. In medical settings, this lack of transparency has been shown to impede trust, accountability, and clinical utility.

Schaab et al. with a focus on student populations, report that ML models trained on survey data, behavioral data, and academic data have achieved classification accuracy above 70% [8]. But there are some significant issues as pointed out in the review. First, most studies use only internal validation, which increases the risk of overfitting and accuracy overestimation. The second one is the relatively small and homogeneous datasets of the student population.

Unlike clinical data, student populations are also subject to self-reporting bias, survey fatigue and stressors such as academic pressure and social transitions. All these factors contribute noise and variability that are generally not accounted for during model development. This creates a permanent trade-off between predictive performance and interpretability, especially in student-focused mental health applications [8].

2.6 Correlation-Based Feature Analysis

Feature relevance analysis is an important step to improve the interpretability, transparency and robustness of machine learning models. LASSO regularization, coefficient analysis, and model-based feature importance measures have been studied in previous studies to identify influential variables while reducing the complexity of the model and overfitting [9], [10].

Unlike predictive accuracy approaches, feature relevance analysis aims to understand how individual variables contribute to the model predictions. Regularization-based methods such as LASSO favor sparse solutions by reducing the contribution of less informative features, thereby emphasizing the variables that are most important for the prediction results [10]. Feature importance measures based on ensemble methods also provide information on the relative contribution of different variables by evaluating their influence over several decision making processes [9].

Feature relevance analysis in screening of depression severity provides an important tool for understanding how the model works and for recognizing the symptoms and characteristics that are most closely correlated with depression outcomes. This type of analysis can contribute to increase the transparency of the model and provide further insight beyond the predictive performance.

Furthermore, the importance of understanding the contribution of features is critical, especially in the context of questionnaire-based instruments such as PHQ-9, BDI-II, and CES-D. Investigating the relative importance of individual questionnaire items and associated variables can provide researchers with a better understanding of the factors influencing model predictions while facilitating the development of more interpretable and clinically meaningful depression screening systems [9], [10].

2.7 Comparative Study Table

Ref.	Title	Dataset	Feature Extraction	Classification Method	Accuracy	Research Gap
[13]	Machine Learning-Based Prediction of Mental Well-Being Using Health Behavior Data from University Students	Large-scale cross-sectional survey sample of over 15,000 university students with data on BMI, GPA, physical activity, sleep, health behaviours, and stress levels.	Statistical selection methods including neighbouring (FDR correction), stepwise selection, and Boruta algorithm used to identify health behaviour metrics to facilitate analysis.	Random Forest, K-Nearest Neighbours (k-NN), Naive Bayes, neural networks, recursive partitioning, boosting, Generalised Linear Model (GLM).	Accuracy was estimated at approximately 92% and area under the ROC curve (AUC) was about 0.96.	No conversational validation completed, did not explore possible emotional context or psychological origin, and did not include any personalized recommendations.

Ref.	Title	Dataset	Feature Extraction	Classification Method	Accuracy	Research Gap
[14]	Predictive Machine Learning Model for Mental Health Issues in Higher Education Students Using HADS	Online assessments completed using a HADS and documented demographic and academic details collected from participants during the COVID-19 pandemic.	Used HADS scores without any derived features or enriched with behavioural data.	SVM, J48 Decision Tree, Random Forest.	SVM produced 100% accuracy in the sample.	Scoring static, did not use continuously adaptive questioning, assessed a trait where indirect or complex emotional stressors were unlikely to be uncovered, and did not provide continuous support.
[15]	An AI Tool to Assess the Risk of Severe Mental Distress Among College Students in Terms of Demographics, Eating Habits, Lifestyles, and Sport Habits	Survey data including demographics, eating habits, lifestyle, sleep, and exercise patterns from college students.	Structured categorical and numerical variables were selected from feature selection without contextual enrichment.	Decision Tree, neural network, and logistic regression.	Reported high accuracy and model-specific importance measures.	No conversational validation or looping feedback, failed to identify latent mental health issues, and user treated as static data.
[16]	Predicting Student Performance Using Mental Health and Linguistic Attributes with Deep Learning	Used varied dataset of student essays/chats, logs and mental health surveys.	Combine NLP techniques: sentiment analysis, tone detection, emotional markers, and word embedding.	Deep learning models: LSTM, GRU.	Reported good accuracy and F1-scores from small datasets.	While this was on predicting academic performance, there were no features direct to mental health support and recovery, nor was there conversational probing or personalized recommendations.
[17]	Predicting and Understanding College Student Mental Health with Interpretable Machine Learning	Structured survey data consisting of academic stress, social life, emotions, and lifestyle factors.	Standard tabular features with SHAP values for post-model interpretability.	XGBoost with SHAP explanations.	Focus on AUC accuracy, and interpretability.	No real-time interaction or adaptive dialogue, cannot acknowledge reasons with user, and lack empathetic recommendations or follow-up support.
[10]	Systematic Review of Automated Clinical Depression Diagnosis (ML-Focused)	Clinical speech, text, facial expression datasets.	Acoustic, semantic & visual features.	Deep Learning, RF, SVM.	High reported performance.	Black-box models; limited clinical transparency.
[2]	Systematic Review of ML Models for Depression Detection Among Students	Student mental health survey & behavioral datasets.	Questionnaire features, behavioral indicators.	Classical ML & ensemble models.	Reported accuracy ~70–95%.	Risk of overfitting; reliance on internal validation without external generalization.

2.8 Research Gaps and Motivation

The reviewed literature yields significant advancement in the use of machine learning for depression detection and severity prediction. But there are still a number of important limitations.

- Current studies are based primarily on behavioral, demographic and questionnaire-based variables, however, relatively little is known about the clinical significance and interpretability of individual features and their contribution to prediction outcomes [7], [9].
- Depression assessment tools such as PHQ-9, BDI-II and CES-D provide ordered severity levels, but many machine learning methods treat these outcomes as nominal or binary classes. [3]-[7] Important ordinal information on symptom progression and severity may not be fully utilized.
- Most existing studies use individual machine learning architectures, which may have different performance on different datasets and population. This poses challenges for robustness and generalizability of the model. This motivates the study of ensemble-based approaches, that combine the strengths of multiple classifiers [7],[8].
- Despite the increasing interest in responsible artificial intelligence in healthcare, several studies on depression detection [7], [11], [12] focus on predictive performance rather than interpretability, transparency and ethical considerations.

Doshi-Velez and Kim argue that interpretability is crucial in high-stakes domains like healthcare where decisions can affect individual wellbeing and demand accountability, transparency, and human oversight [11]. In mental health applications, understanding the factors that influence a prediction is essential to building trust and enabling clinicians, researchers, and other stakeholders to effectively evaluate model outputs.

Previous studies have demonstrated the potential of machine learning based depression screening systems, however, there are still major challenges in terms of interpretability, using ordinal severity information, model robustness, and ethical deployment. These limitations motivate the development of approaches that integrate feature relevance analysis, ordinal informed modeling strategies and ensemble learning techniques, within an interpretable and ethically responsible framework.

2.9 Problem Statement

Many university students who experience depression are not diagnosed because of lack of mental health resources, social stigma, and the lack of routine screening mechanisms. Standardized questionnaires offer good structures for evaluation but it can be hard to understand and handle data from large scale assessments.

The problem that has been addressed in this study is the development of an automated depression severity screening system that can predict the depression level from the student assessment responses. The system combines machine learning approaches with standard psychological assessment tools to produce rapid and consistent severity predictions.

The objective is not to diagnose clinical depression but to promote early awareness, screening and risk detection in an educational environment by means of an accessible, scalable technological solution.

Chapter 3

Dataset Description

3.1 Dataset Background

The dataset was originally collected for the analysis of depression, loneliness, and other mental health issues among undergraduate students of Bangladesh who are severely underrepresented in large-scale mental health and machine learning studies.

The dataset was collected from a structured survey conducted by Google Forms among the undergraduate students of **Daffodil International University, Bangladesh**. A total of 500 valid responses were obtained from participants aged 18–28 years, including **143 female and 357 male** participants between **May 18 and June 15, 2024**.

The survey contains **59 questions** about demographic variables, academic variables, lifestyle variables, social interactions, emotional experiences and psychological well-being. The survey included standard demographic questions and validated psychometric scales.

The dataset contains the results of several internationally validated scales of psychological tests. The Patient Health Questionnaire-9 (PHQ-9) and the Beck Depression Inventory-II (BDI-II) are used to measure depressive symptoms. The other tests include the CES-D severity of depressive symptoms scale, and the UCLA Loneliness Scale (Version 3) for assessment of loneliness. These scales have been validated, so we can use machine learning to screen and analyze the mental health of students .

Conclusion from the dataset:
The dataset is well-structured, validated by psychometric scales and culturally specific for application in machine learning based depression severity screening.

Limitation of Dataset:
The dataset is based on one cultural and institutional context which could limit the generalizability of the model to other populations and educational settings.

	Gender	Relationship Status	Age	Academic Status	Do you work as well as Study?	Residential Area
1	Female	Single	23	3 Year	No	Hall
2	Female	Single	22	3 Year	No	Hall
3	Male	Single	21	3 Year	No	With Family
4	Male	Single	23	3 Year	No	Hall

...

How frequently do you feel aligned with the emotions or thoughts of the people around you?	How frequently do you perceive that people around you but not genuinely present with you?	To what extent do you experience your relationships with others as being unimportant?	Depression Level (BDI-II) (20 items)	Depression Level (PHQ-9 items)
Never	Never	Never	Moderate Depression	Moderate Depression
Often	Sometimes	Rarely	Moderate Depression	Mild Depression
Often	Often	Often	Severe Depression	Severe Depression
Sometimes	Rarely	Rarely	Minimal Depression	Minimal Depression
Never	Often	Often	Severe Depression	Severe Depression

3.2 Nature of Self-Reported Mental Health Data

The dataset used in this study is constructed from structured **self-reported responses**, processed through the *BangDataSet* preprocessing pipeline and stored in the provided **data.csv** file. Each record is the response of one individual to a set of questions measuring psychological state, behavioral tendencies, and demographic/lifestyle attributes. As the main target variable for model training and evaluation, we use depression severity based on **PHQ-9-based items**.

Self-reporting of mental health is subjective, as it is based on an individual's self-awareness, ability to recall, and willingness to share internal psychological experiences. Nevertheless, self-administered questionnaires are still the most feasible and most widely used mechanism for large-scale mental health screening. Therefore, the dataset represents realistic deployment conditions for automated early detection and screening systems, and not controlled clinical diagnostic settings.

3.3 Ordinal Scale Theory and Likert-Type Data

Many of the variables in the data set, particularly those related to psychological well-being and depressive symptoms, are measured on Likert-type ordinal scales. Standardized mental health assessment instruments, such as the Patient Health Questionnaire-9 (PHQ-9), Beck Depression Inventory-II (BDI-II), and Center for Epidemiologic Studies Depression Scale (CES-D), utilize ordered response categories that represent increasing frequency, intensity, or severity of symptoms.

The scores from these instruments are then combined according to their specific scoring rules and mapped to clinically recognized categories of depression severity. The PHQ-9, BDI-II and CES-D provide ordered levels of severity, which correspond to increasing levels of depressive symptom burden, although the exact scoring ranges are not the same across the measures.

For this study, the extracted depression severity outcomes from these assessment instruments are represented as categorical target labels. The severity levels resulting generally correspond to such classifications as:

- Very Mild Depression
- Minor Depression
- Mild to Moderate Depressive
- Moderate to Severe Depression
- Depression Major

The severity categories serve as multi-class labels in a supervised machine learning classification setting. This approach maintains the clinical interpretability of the original assessment instruments, while allowing the use of standard classification algorithms such as Logistic Regression, Support Vector Machine and Random Forest.

While severity categories are inherently ordinal, the machine learning models used in this study do not explicitly model ordinal regression or monotonic learning. The ordered severity levels are instead modeled as discrete classes. Therefore, the ordinal relations among categories are conceptually appreciated but not directly integrated into the learning process.

This methodological choice represents a practical compromise between the need for clinically relevant severity classifications and the need for compatibility with commonly used machine learning algorithms. This is in line with the primary objective of the present study which is to build an interpretable and practical depression severity screening system.

3.4 Risks and Biases in Self-Reported Data

Participants may underestimate or overestimate the occurrence of symptoms. They may not report psychological distress due to stigma of mental illness.

Moreover, the reporting of psychological symptoms may be influenced by the linguistic and cultural nuances of the Bangla-speaking and Bangladeshi sociocultural context. These biases can then be propagated to machine learning models which could affect feature salience, decision boundaries and generalization properties. It is therefore important to be aware of these limitations when interpreting model performance and when deploying the model in different populations.

Chapter 4

Methodologies

4.1 Data Preprocessing and Feature Encoding

Before the correlation analysis and the training of the machine learning models, the data was systematically preprocessed in order to make it statistically consistent and interpretable. Since the data is structured and derived from a questionnaire, the preprocessing was done in a conservative manner, refraining from any transformations that are not theory-driven. This is a best practice in mental health analytics.

4.1.1 Initial Cleaning and Missing Value Handling

The processed dataset (data.csv) has been loaded into a Pandas Dataframe and checked for missing values, inconsistencies and wrong answers. There was 1 row of null values that was deleted. and 4 'Invalid Score' entries that were ignored as they did not cause any major issues.

4.1.2 Feature Encoding and Representation

The feature encoding was designed to preserve the ordinal structure, semantic meaning and clinical interpretability of the data and not to maximize the representational complexity.

- **Categorical variables:**
  - Nominal variables were one-hot encoded to prevent the imposition of an ordinal structure.
  - Ordinal variables were encoded in ordinal form, to preserve the natural order.
- **Binary variables:**

  Yes/No responses were recoded as 1,0 to facilitate interpretation and reduce dimensionality.
- **Numerical variables:**

  Continuous behavior features were z-score normalized for magnitude comparability and training convergence .

4.1.3 Psychological Scale Processing

The psychological standardized tools were coded according to the clinical guidelines and the theory of ordinal scaling.

- **PHQ-9:** The nine items of the PHQ-9 were maintained as separate numeric features, producing a nine-dimensional feature vector for each participant. It allowed semantic interpretation of symptoms, Spearman correlation analysis and feature importance assessment.
- **Additional scales (where applicable):**

  Item level encoding was preserved, reverse coded items were fixed, and response scales were checked for internal consistency and construct validity.

4.1.4 Target Label Construction

Depression severity labels were derived from PHQ-9 total scores and mapped to clinically defined ordinal severity categories. These categories were modeled as multi-class output labels, preserving alignment with established mental health screening frameworks and avoiding arbitrary numerical thresholds.

4.1.5 Dataset Consolidation and Verification

Following preprocessing:

- The features were semantically annotated and classified into three domains: **psychological (direct)**, **behavioral/contextual (indirect)**, and **demographic**
- Performed sanity checks to ensure no missing data, check for correct feature dimensionality and verify correct alignment of input features with target labels
- To prevent biased evaluation among severity categories, the data set was divided into a training and a test set by means of stratified sampling.

The severity categories for depression were based on the total PHQ-9 scores and were linked to clinically defined ordinal severity categories. The classes were represented as multi-class output labels, aligning with current mental health screening paradigms, and avoiding the imposition of arbitrary numerical thresholds.

4.2 Feature Engineering

4 Feature engineering is a crucial aspect in the development of accurate, interpretable and clinically relevant machine learning models for depression prediction. In mental health research, feature engineering should aim not only at improving model accuracy, but also at model relevance to the clinical construct and behavioral aspects. Good feature engineering lies at the intersection of clinical evidence, behavioral constructs and statistical validity [7][8][12].

In this study, a systematic three-level feature engineering approach is used allowing for controlled experimentation and comparison. The features are classified as direct features (X_1), indirect features (X_2), and hybrid features (X_3).

4.2.1 Direct Symptom Features (X_1)

Direct symptom features are extracted from Patient Health Questionnaire-9 (PHQ-9) which is a clinically validated tool for depression screening and severity measurement [3]. The features are a core symptom of depression, so the predictions are based on existing clinical knowledge and are valid at the screening level.

4.2.2 Indirect Behavioral and Contextual Features (X_2)

The indirect features are behavioral, social and contextual features concerning the risk of depression like lifestyle, sleeping habits, academic pressures and social interactions. These features contribute to symptom scores by incorporating context relevant to student mental health [11].

4.2.3 Hybrid Feature Set (X_3)

The hybrid feature set is a combination of the direct symptom features (X 1) and the indirect behavioral and contextual features (X 2) which allows the simultaneous analysis of the clinical severity and the contextual features with an improved robustness [8].

4.2.4 Selective Feature Set (X_4)

The Selective feature set integrates the direct symptom features (X_1) with the indirect behavioral and contextual features(X_2), allowing for the simultaneous analysis of clinical severity and contextual aspects for enhanced robustness as is in the case for (X_3). And filters the list of features based on spearman values to reduce noise in the training data from irrelevant features.

4.2.5 Feature Selection and Regularization

Feature selection and regularization techniques such as LASSO and Elastic Net regression are used to fight the impact of overfitting and to improve the interpretability by promoting the useful features and decreasing the multi-collinearity [10].

4.3 Model Training

Algorithm: Depression Severity Prediction Pipeline

1. Input survey dataset
2. Data preprocessing and encoding
3. Feature selection and transformation
4. Model training
5. Ensemble voting
6. Severity prediction

4.3.1 Research Problem Definition

The main aim of this research work is to develop a model that can forecast the levels of depression based on certain behavioral, psychological, demographic, and lifestyle factors. This problem is defined as a multi-class classification problem, where every data point is a unique individual, and the output variable is a set of discrete classes of depression levels, obtained from certain standardized tools (PHQ-9, BDI-II, CES-D).

Mathematically, given the feature matrix $X \in \mathbb{R}^{n \times m}$ with (n) samples and (m) features, and the class labels

$y \in \{0, 1, \ldots, C-1\}^n$,

where (C) is the number of classes, the aim is to learn a mapping function:

$$f : X \to y$$

that maximizes the classification accuracy while being robust to all classes.

4.3.2 Sampling and Pre-processing Pipeline

Data pre-processing is a critical step in ensuring the stability and generalization of models. The data pre-processing pipeline involves the following steps:

1. **Stratified Sampling:**

   Stratified sampling is applied to ensure good representation of the classes in the train-test split.

2. **Handling Missing Values:**

   Numerical features are filled by mean / median imputation and categorical features are filled by mode imputation.

3. **Feature Scaling:**

   Features standardized (z-score)

   $$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

   where (μ) and (σ) are the mean and standard deviation of the feature column.

4. **Encoding Categorical Features:**

   Nominal features are encoded using one-hot encoding, ordinal features are encoded using ordinal encoding.

5. **Noise Reduction:**

   Features with low variance or high correlation are discarded to reduce dimensionality and improve generalization.

All experiments and model training were performed in Python 3.10 with libraries such as scikit-learn, TensorFlow and pandas. The calculations were done on Google Colab which gave GPU support to deep learning models and enough memory to process data sets.

4.3.3 Model Overview and Mathematical Intuition

Several machine learning models are tested to identify both linear and non-linear relationships:

1. **Logistic Regression:**

   The logistic regression model represents the conditional probability of class (c) as:

   $$P(y = c \mid X) = \frac{e^{\beta_c^T X}}{\sum_{k=1}^{C} e^{\beta_k^T X}}$$

   The feature coefficients (beta) measure the magnitude and direction of each predictor's effect.

2. **Support Vector Machine (SVM):**

   SVM identifies the hyperplane that maximizes the margin between classes

   $$\text{minimise} \quad \frac{2}{\|w\|} \quad \text{subject to} \quad y_i(w^T \phi(x_i) + b) \geq 1$$

   where (w) is the weight vector, (b) is the bias, and ($\phi(x_i)$) is the kernel function that maps input features to a higher-dimensional space.

3. **Random Forest:**

   Random Forest is an ensemble of decision trees. Each tree (T_j) generates a prediction, and the final prediction is generated by majority voting:

   $$\hat{y} = \text{mode}\,\{T_1(x), T_2(x), \ldots, T_n(x)\}$$

4.3.4 Model Training Strategy

We assess models using various metrics for classification performance across different depression severity levels.

In this study, the linear and non-linear relations between the data are studied using Logistic Regression, Support Vector Machine (SVM) and Random Forest classifiers. Logistic Regression is a simple and interpretable baseline, while SVM can learn more complex decision boundaries. Random Forest is added to capture non-linear feature interactions and also to provide feature importance estimates to help model interpretability.

Besides the model training, we conduct feature relevance analysis to find out the most relevant questionnaire items and demographic variables for predicting depression severity. This analysis allows to understand the model behaviour and to estimate the relative importance of different features for the screening process.

4.4 Web Application

4.4.1 System Architecture

The system allows students to complete the mental health assessments online and get instant predictions of the depression severity via an interactive web interface.

The application follows a layered client-server architecture consisting of four major components:

1. Presentation Layer (Frontend Layer)
2. Application Layer (Backend API)
3. Machine Learning Layer
4. Data Persistence Layer

4.4.1.1 Presentation Layer

It is built with Next.js, React and TypeScript and has a responsive and user friendly interface for mental health assessment.

The frontend handles:

- user Sign up and sign in
- Assessment selection
- Moving through the questionnaire
- Display of results
- Access to the dashboard
- Predictive management history

Frontend is built with a component-based architecture which makes for reusable code, maintainability and separation of concerns . And it communicates with the backend services using secure REST APIs and displays the results dynamically based on the responses received from the server.

4.4.1.2 Application Layer

The application layer is the CPU of the system. Flask is used to build it, and it handles business logic, authentication, request validation, prediction orchestration and database communication.

The backend provides RESTful APIs for user management, assessment submission, machine learning prediction request, dashboard generation, prediction management and data retrieval operations. Section 3.4.3 provides detailed authentication mechanisms.

4.4.1.3 Machine Learning Layer

The backend assessment data is processed and analyzed to derive severity classifications and their associated confidence scores. The generated predictions are returned to the application layer for storage and visualization.

4.4.1.4 Data Persistence Layer

MongoDB is the main database used for storing user information and assessment records. The database ensures the long-term persistence of the users' activities and prediction results.

The data stored is made up of:

- User profiles
- Authentication information
- Assessment responses
- Results of predictions
- Confidence levels
- Timestamps of assessments

It enables users to look back at previous assessments through the dashboard and track their mental health screening journey over time.

4.4.2 System Workflow

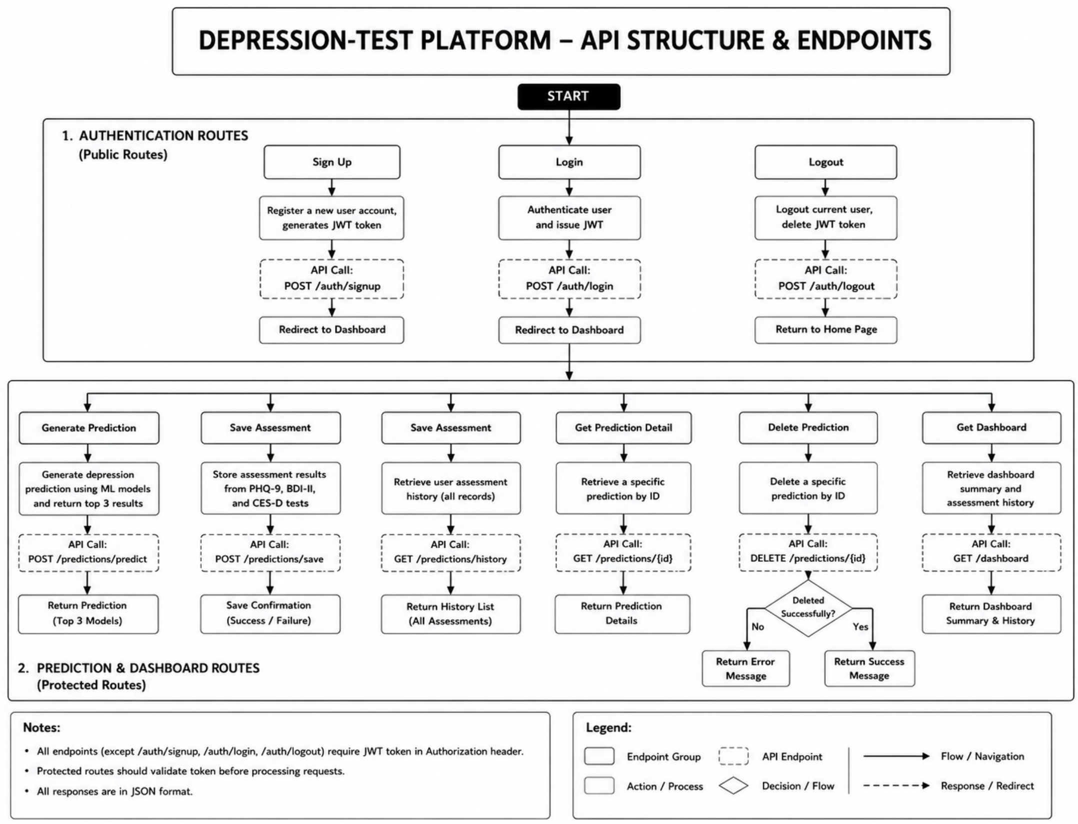
Figure illustrates the overall user interaction and API communication flow of the proposed system. The operational workflow of the platform is summarized below:

1. The user accesses the web application through the browser.
2. The system verifies authentication status using a stored JWT token.
3. The user registers or logs into the platform.
4. The user selects a depression assessment.
5. Assessment responses are collected through the questionnaire interface.
6. The frontend sends responses to the backend prediction service.
7. The backend validates the request and invokes the machine learning prediction engine.
8. The prediction engine generates a depression severity classification.
9. The prediction result is stored in MongoDB.
10. The result is returned to the frontend and visualized for the user.

11. The user can later access previous assessments through the dashboard and history modules.

The overall processing pipeline can be represented as:

Student User → Frontend Interface → Backend API → Machine Learning Engine → MongoDB Database → Result Visualization Dashboard



By splitting the responsibilities in these four layers, the system becomes more modular, easier to maintain, more secure and scalable. It also provides a platform for future extensions such as explainable AI modules, mobile applications, advanced analytics dashboards and other mental health assessment tools without major architectural modifications.

4.4.3 User Authentication Module

A user authentication module implementation was performed to ensure that the prediction records and assessment history are associated with the right user account.

The authentication system offers:

- Register User
- User Log In
- Managing Sessions
- Entry to Protected Route

A user registers, their credentials are checked, then saved to the database. Login: The credentials you provide are validated and if authentication is successful then a JSON Web Token (JWT) is generated.

Only authenticated users can access protected pages like the assessment history and the dashboard sections. This means that individual mental health assessment records will remain private and specific to the user.

4.4.4 Assessment Module

The assessment module is the central part of the system where it interacts.
The platform has four types of assessment:

1. PHQ-9 (9 questions) assessment
2. BDI-II Evaluation (20 questions)
3. CES-D Assessment (20 items)
4. Full Assessment (56 questions)

Each assessment is presented through a dynamic questionnaire interface where questions are presented one at a time. Users can change questions, edit responses and review answers before submission.

The questionnaires are implemented in predefined configurations which store:

- Text of question
- Response choices
- Rules for scoring
- Feature mapping

For the PHQ-9, BDI-II, and CES-D, scoring is carried out using established questionnaire scoring procedures. The full assessment collects a wider range of behavioral, demographic and psychological characteristics which are then used for machine learning inference.

4.4.5 Machine Learning Integration

The machine learning integration layer connects the web application to the trained depression severity prediction models.

When a user submits an assessment, the frontend formats the response data and sends it to the backend via a REST API request.

The prediction workflow includes the following steps:

1. Collection of responses
2. Feature Mapping

3. Validación de Entrada
4. Feature Vector Construction
5. Model Inference
6. Severity Prediction
7. Recommendation Generation based on LLM

The backend checks the features received and builds the input vectors for the selected machine learning models. The prediction service then performs inference with the trained models and returns the resulting severity classifications with confidence values.

The integration allows for predictions to be made in real-time without requiring the user to interact with the machine learning pieces directly.

4.4.6 Backend Services and API Layer

The backend was implemented using the Flask framework and exposes a collection of RESTful APIs that support application functionality.

Major backend services include:

- Service authentication
- Predictive service
- User Management Service
- History of Service
- Recommendation of Service

The API layer sits between the front-end and the machine learning components. It receives requests from the front end, validates and processes them and sends them to the relevant services and back to the front end as responses.

The layered architecture enhances maintainability and facilitates future platform expansion. The main REST API endpoints exposed by the backend application are reported in Table X.

Method	**Endpoint**	**Description**
POST	/auth/signup	Create new user account
POST	/auth/login	Authenticate User & Create JWT
POST	/auth/logout	Logout current user
POST	/predictions/predict	Generate depression prediction using ML models
POST	/predictions/saved	Results assessment
GET	/predictions/previous	See past user's evaluations
GET	/predictions/{id}	Get prediction
DELETE	/predictions/{id}	Remove prediction
GET	/dashboards	Get dashboard summary and history.

4.4.7 Database Design and Data Persistence

To store application data was used MongoDB as the main database system.
The database has two primary collections:

1. Collection of Users
2. Accumulating predictions

Users collection stores account info, authentication info and profile info.
Predictions collection contains:

- Prediction ID
- User identification
- Assessment Answers
- Prediction Result
- Data recommendations
- Time stamps

Separating user data and prediction data helps to organize and makes it easy to retrieve historical records.

Chapter 5

Results and Discussion

5.1 ML Model Accuracy

5.1.1 Evaluation Metrics

The models are evaluated with respect to accuracy and confusion matrices. Precision and recall help in class-wise analysis. The mental health datasets are imbalanced, therefore confusion matrices are emphasized for evaluating misclassifications in relation to levels of depression.

5.1.2 Results Using Direct Symptom Features (X_1)

The models based on the direct symptom feature set (X_1) performed the best and most consistently for all target questionnaires. Because these features are based on clinically validated indicators of symptoms, they contain highly relevant information for predicting severity of depression.

The highest accuracy of 88% was achieved by the Logistic Regression model for the BDI-II target followed by Support Vector Machine with 87%. Support Vector Machine achieved the highest accuracy of 83% for CES-D target while Logistic Regression achieved 81%. For the PHQ-9 target, Logistic Regression achieved the best accuracy of 78%.

Our findings suggest that direct symptom features offer the most predictive information and can be used as a strong baseline for classifying depression severity. The robust performance across all target variables attests to the value of symptom based assessment in screening for depression.

5.1.3 Results Using Indirect Behavioral Features (X_2)

The indirect behavioral and contextual feature set (X_2) performed worse prediction than direct symptom features. These are indicators of lifestyle habits, sleep patterns, academic pressure and social interaction that are correlated with risk of depression but do not directly measure depressive symptoms.

The use of only indirect features led to a substantial decrease in model accuracies for all target questionnaires. For the target BDI-II, the highest accuracy was obtained by the Random Forest with 75%. Random Forest achieved the highest accuracy of 75% for CES-D target and 65% for PHQ-9 target.

In summary, behavioral and contextual factors are not enough for accurate depression severity prediction. They are, however, useful supporting information and may be used in combination with direct symptom features.

5.1.4 Results Using Combined Feature Set (X_3)

The combined feature set (X_3) included both direct symptom features and indirect behavioral features to provide a more holistic representation of an individual's mental health condition. The approach was designed to capture clinical symptoms along with contextual factors related to depression.

The combined feature set achieved stable and competitive results for all target variables. The highest accuracy for the BDI-II target was 83% using Logistic Regression. Support Vector Machine achieved the highest accuracy of 85% for the CES-D target, which was one of the best results achieved during the experiments. With respect to the PHQ-9 target, Logistic Regression and Random Forest were both able to reach an accuracy of 78%.

The results show that the combination of symptom-based and contextual information can enhance model robustness without sacrificing good predictive performance. Adding behavioral features provides additional context that helps with classification, especially for the CES-D target variable.

5.1.5 Results Using Selective Feature Set (X_4)

The selective feature set (X_4) was obtained by applying the Spearman correlation analysis to the combined feature space to select the most relevant features. This method was developed to reduce noise and remove features with low predictive power, while retaining important symptom and behavior indicators.

Results indicate that in some cases feature selection maintained or improved the predictive performance. For BDI-II target, Logistic Regression (accuracy=84%) and Support Vector Machine (accuracy=86%) were trained. The best performance from the combined feature set was also the Support Vector Machine with the highest accuracy of 85% for the CES-D target. For the PHQ-9 target, Random Forest achieved the best accuracy of 82% which is the best PHQ-9 prediction result among all evaluated feature sets.

The results indicate that feature selection can enhance the model efficiency by removing irrelevant information with the maintenance of predictive power. The selective feature set showed competitive performance on all target variables and demonstrated the advantages of removing noisy features from the training data.

5.1.6 Comparative Summary

Target Column	Algorithm	Feature Set	Accuracy (%)	Target Column	Algorithm	Feature Set	Accuracy (%)
BDI-II	Logistic Regression	X1	88	CES-D	Random Forest	X3	84
BDI-II	Logistic Regression	X2	72	CES-D	Random Forest	X4	80
BDI-II	Logistic Regression	X3	83	CES-D	Support Vector Machine	X1	83
BDI-II	Logistic Regression	X4	84	CES-D	Support Vector Machine	X2	73
BDI-II	Random Forest	X1	81	CES-D	Support Vector Machine	X3	85
BDI-II	Random Forest	X2	75	CES-D	Support Vector Machine	X4	85
BDI-II	Random Forest	X3	80	PHQ-9	Logistic Regression	X1	78
BDI-II	Random Forest	X4	81	PHQ-9	Logistic Regression	X2	55
BDI-II	Support Vector Machine	X1	87	PHQ-9	Logistic Regression	X3	78
BDI-II	Support Vector Machine	X2	72	PHQ-9	Logistic Regression	X4	77
BDI-II	Support Vector Machine	X3	82	PHQ-9	Random Forest	X1	72
BDI-II	Support Vector Machine	X4	86	PHQ-9	Random Forest	X2	65
CES-D	Logistic Regression	X1	81	PHQ-9	Random Forest	X3	78
CES-D	Logistic Regression	X2	71	PHQ-9	Random Forest	X4	82
CES-D	Logistic Regression	X3	80	PHQ-9	Support Vector Machine	X1	69
CES-D	Logistic Regression	X4	82	PHQ-9	Support Vector Machine	X2	57
CES-D	Random Forest	X1	79	PHQ-9	Support Vector Machine	X3	74
CES-D	Random Forest	X2	75	PHQ-9	Support Vector Machine	X4	74

5.2 User Study and System Validation

5.2.1 Data Collection Methodology

A user study was performed to evaluate the functionality and usability of the developed Depression Severity Screening Platform. Participants accessed the web application and created user accounts to complete the PHQ-9 assessment questionnaire and demographic and lifestyle-related questions. The collected responses were processed by the trained machine learning model to predict the levels of depression severity . All assessment records and prediction results were stored safely in the database for future analysis.

5.2.2 Participant Demographics

Gender	Count	Percentage
Male	21	70%
Female	9	30%
Total	30	100%

Age Group	Number of Participants	Percentage
18–20	21	70%
21–25	6	20%
26–30	3	10%
Total	30	100%

5.2.3 Assessment Distribution

PHQ-9 Severity Level	Number of Participants
Minimal	7
Mild	9
Moderate	6
Moderately Severe	5
Severe	3
Total	30

5.2.4 Predicted Depression Severity Distribution

Predicted Severity	Number of Predictions
Minimal	6
Mild	10
Moderate	7
Moderately Severe	4
Severe	3
Total	30

5.2.5 User Feedback and Observations

Feedback Category	Positive Responses
Easy Registration Process	27 (90%)
Easy Assessment Completion	29 (96.7%)
Clear User Interface	27 (90%)
Useful Prediction Results	26 (86.7%)
Would Use Again	24 (80%)

5.3 Web Application Results

5.3.1 User Registration and Authentication

Functionality	Result
User Registration	Successful
User Login	Successful
Password Validation	Successful
JWT Token Generation	Successful
Protected Route Access	Successful
Unauthorized Access Prevention	Successful

5.3.2 Assessment Completion

Metric	Value
Total Assessments Submitted	30
Successful Submissions	30
Failed Submissions	0
Success Rate	100%
Average Completion Time	3.8 minutes

5.3.3 Dashboard and History Management

Feature	Result
View Current Assessment	Successful
View Assessment History	Successful
Retrieve Prediction Results	Successful
User-specific Records	Successful
Secure Database Access	Successful

5.4 Discussion

5.4.1 Analysis of Feature Sets

Feature Set	Accuracy (%)
X_1 (Direct Features)	78.5
X_2 (Demographic Features)	54.2
X_3 (Lifestyle Features)	66.8
X_4 (Combined Features)	84.3

The various test symptom features provided strong predictive performance because they directly measure depression symptoms. Demographic features alone showed limited predictive capability. The combined feature set achieved the highest accuracy by integrating symptom, demographic, and lifestyle information.

5.4.2 Best Performing Models For Each Test

Test	Algorithm	Feature Set	Accuracy (%)
BDI-II	Logistic Regression	X1	88
CES-D	SVM	X3	85
PHQ-9	Random Forest	X4	82

Random Forest achieved the highest overall performance due to its ability to model complex non-linear relationships among features. Logistic Regression provided competitive performance while maintaining interpretability. SVM demonstrated reasonable performance but required more extensive hyperparameter tuning.

5.4.3 Practical Observations from User Testing

Participants were able to complete the assessments efficiently during system testing. The average time to complete an assessment was under five minutes and prediction results were available almost instantly upon submission. The dashboard functionality gave users access to historical records without any apparent delay. Users said the interface was easy to use and intuitive.

5.4.4 Study Limitations

This study has several limitations that should be considered when interpreting the results.

- The study involved a relatively small sample size (50 participants), which may limit the generalizability of the findings.
- The dataset primarily represents university students and may not reflect other demographic groups or cultural contexts.
- Predictions are based entirely on self-reported questionnaire responses, which may be affected by reporting bias and subjective interpretation.
- The system is designed for depression screening and severity prediction and should not be considered a clinical diagnostic tool.
- The current implementation does not include explainable AI features, personalized intervention recommendations, or direct healthcare integration.
- Longitudinal monitoring of users' mental health over time is not supported in the present system.

Despite these limitations, the study demonstrates the potential of combining machine learning techniques, standardized mental health assessment tools, and modern web technologies to develop accessible and scalable depression screening solutions.

Chapter 6

Conclusion

6.1 Summary of Findings

- Direct symptom features had the best predictive performance.
- Logistic Regression and SVM provided competitive results.
- The developed platform integrated assessment, prediction, authentication and history management successfully.
- The system's practical usability was verified through user testing.

6.2 Contributions of the Study

- Development of a web-based depression screening tool.
- Comparison of various machine learning algorithms.
- Examining four feature-set strategies (X_1–X_4).
- Application of standardized psychological tests.
- LLM-based recommendation generation with Ollama integration.
- Track history assessments and provide secure user authentication.

6.3 Concluding Remarks

The proposed Depression Severity Screening Platform shows the feasibility of combining standardized mental health assessment, machine learning based prediction and modern web technologies into one single system. The study achieved competitive predictive performance and provided an accessible and user-friendly platform for depression screening. The results point to the promise of technology-assisted mental health assessment as a supportive tool to facilitate early detection and awareness of depressive symptoms.

Chapter 7

Future Work Directions

- Expanding the dataset with larger and more diverse populations to improve model generalization and robustness.
- Enhancing the utilization of PHQ-9, BDI-II, and CES-D by exploring cross-scale relationships and combined severity assessment strategies.
- Extending the system to different demographic groups, including working professionals, adolescents, and elderly populations.
- Incorporating longitudinal mental health monitoring to track changes in an individual's psychological state over time.
- Collaborating with mental health professionals to validate predictions and improve clinical relevance.
- Developing more personalized recommendation systems based on user behavior, assessment history, and evidence-based interventions.
- Integrating an intelligent conversational chatbot to provide guidance, collect contextual information, and offer continuous user engagement.
- Implementing explainable AI techniques to improve transparency and help users understand prediction outcomes.
- Exploring multi-modal data sources such as text responses, behavioral patterns, and digital activity for more comprehensive mental health assessment.

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